

EXPLORATORY DATA ANALYTICS (EDA) REPORT

The E-commerce Sales Analysis was carried out using Exploratory Data Analysis (EDA) techniques in Excel to identify trends, detect unexpected pattern and data , and generate actionable business insights.

Descriptive statistics were performed on the numerical columns using both basic Excel functions and the Data Analysis Tool Pak(Descriptive Statistics). Both methods produced consistent results, revealing that customers purchased an average of 2.95 units per transaction, with unit prices averaging approximately \$356 and ranging from \$11 to \$700. Customers had an average of 5.5 items in their carts, while the average transaction value was \$1,054, with some exceptionally high-value transactions indicating the presence of outliers. These findings provide an overall understanding of the dataset's distribution and purchasing patterns.

Product Sales Insights and Conditional Formatting

Pivot tables were created from the cleaned dataset to analyze sales performance across products, payment methods, order statuses, and marketing channels. The analysis showed that Chairs, Printers, and Laptops generated the highest revenue, while Phones recorded the lowest. Credit Card and Online payments contributed the most revenue, although Cash had a comparable number of orders. Order statuses were relatively evenly distributed, with notable cancellation and return rates that may indicate operational or customer satisfaction challenges. Among marketing channels, Instagram emerged as the most effective source of revenue, followed by Email and Google, whereas Referral generated the lowest sales. These insights highlight key revenue drivers, customer purchasing behavior, and areas for potential business improvement.

Outlier Analysis was conducted using the Interquartile Range (IQR) method to identify unusually high transaction values. Transactions exceeding the calculated upper bound of \$1,751.93 were classified as outliers, with the highest transaction reaching \$3,456. These exceptional values likely reflect bulk purchases or promotional discounts and may significantly influence average revenue calculations.

Conclusion

The analysis revealed that sales are relatively well distributed across product categories and payment methods, while Instagram proved to be the most effective marketing channel for driving revenue. The high number of cancelled and returned orders suggests potential issues with order fulfillment or customer satisfaction that warrant further investigation. Additionally, the presence of high-value outlier transactions indicates bulk purchasing or promotional activity, which can skew average revenue metrics. Overall, customer

purchasing behavior appears consistent, with a few extreme transactions presenting opportunities for more detailed analysis.